

Shadowing Assessment #3

Date of Shadowing: November 25, 2024

Professional Observed: Ms. Ramya and lab staff

Location: Purepath Lab & Therapeutics

Overview:

In my third visit to Purepath Lab & Therapeutics, I observed the process of diagnostic validation, PCR plating, and the detailed setup for urinary tract infection (UTI) and wound infection tests. This shadowing experience allowed me to see both the scientific and procedural rigor required in molecular diagnostics and the intricate setup needed to ensure reliability and biosafety in medical testing.

Main Observations and Procedures:

- **Validation Tests:** I observed how samples left out over time were tested to confirm if they still produced valid and accurate results — a crucial step to ensure reliability and prevent false positives/negatives.
- **Plate Setup for UTI Diagnostics:**
 - **Wash Plate 1:** 700 μ L of Wash Buffer 1
 - **Wash Plate 2:** 700 μ L of Wash Buffer 2
 - **Wash Plate 3:** 750 μ L of 100% ethanol
 - **Elution Plate:** 100 μ L of molecular-grade water
 - **Sample Plate:** 20 μ L of Proteinase K, 200 μ L of sample (vortexed), 2.5 μ L of exogenous control, 500 μ L of bead mixture (with lysis buffer, binding buffer, and magnetic beads)
- **PCR Plating:** I observed preparation for PCR tests for both wound and UTI detection. Samples were combined with a master mix containing Taq polymerase, primers, and buffer, then sealed, centrifuged, and loaded into the PCR machine.

In-Depth Insight: PCR and qPCR

One of the most important procedures I observed was PCR testing, a technique fundamental to molecular biology. Dr. Tilak's lab uses both conventional PCR and real-time PCR (qPCR) depending on the diagnostic need.

Conventional PCR (Polymerase Chain Reaction):

This technique is used to amplify DNA, making millions of copies of a target sequence. It goes through three core steps:

1. **Denaturation:** The double-stranded DNA is heated ($\sim 95^{\circ}\text{C}$) to separate into single strands.
2. **Annealing:** The temperature lowers ($\sim 50\text{--}65^{\circ}\text{C}$) so that short DNA primers can attach to specific sequences.
3. **Extension:** A heat-stable enzyme, Taq polymerase, synthesizes new DNA strands ($\sim 72^{\circ}\text{C}$).

Although this process is highly effective, conventional PCR doesn't provide real-time data — you only see results at the end (often using gel electrophoresis).

Real-Time PCR (qPCR):

Unlike conventional PCR, qPCR allows scientists to track DNA amplification as it happens by using fluorescent dyes or probes that light up during each cycle. The fluorescence level correlates with the amount of DNA produced, offering both detection and quantification. It's especially useful for:

- **Measuring viral loads (like COVID-19 testing)**
- **Gene expression analysis**
- **Determining sample quality and sensitivity**

qPCR provides a curve showing the cycle threshold (Ct value) — a lower Ct means a higher initial DNA concentration, giving more insight into how much target DNA or RNA was present.

This side-by-side use of both PCR types in a clinical setting emphasized how biomedical engineering bridges molecular biology and diagnostics, offering solutions that are both precise and scalable.

Reflection:

This session deepened my understanding of diagnostic processes and equipment. The clear breakdown of PCR vs. qPCR demonstrated the depth of accuracy and technology needed in molecular testing. From mastering fluid volumes to understanding DNA behavior in elution, I realized how much detail goes into every diagnostic step — and how biomedical innovation drives healthcare forward. I left the lab with a stronger grasp of how engineering, chemistry, and biology intersect in clinical practice.